

## Long-term retinal changes in progressive geographic atrophy.

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Age-related macular degeneration (AMD) is one of the leading causes of blindness with loss of retinal layers over long term. We aim to evaluate these changes in eyes with progressive non-exudative AMD with geographic atrophy (GA). This retrospective study included patients with GA with a minimum of 4 years follow up. Retinal layers on spectral domain optical coherence tomography (SD-OCT) were segmented based on their reflectivity patterns using validated semi-automated segmentation algorithm. The thickness of the segmented retinal layers was measured. Horizontal length of GA at baseline and last follow-up were also measured. Regression analysis was performed to correlate changes in RPE layer thickness with other retinal layers and the length of GA on OCT. A total of 351-line scans including 17 foveal scans showing presence of GA at final visit that is, a total of 2457 retinal layer bands were analyzed. Outer nuclear layer (ONL) ( $p = 0.02$ ), outer segment layers (OSL) ( $p = 0.01$ ), and retinal pigment epithelium (RPE) ( $p = 0.01$ ) showed a statistically significant variation between baseline and final visit. Regression analysis showed the change in ONL ( $r = 0.72$ ;  $p = 0.01$ ) and OSL ( $r = 0.93$ ,  $p < 0.01$ ) correlated significantly with change in RPE thickness whereas rest of the layers failed to show significant correlation. Outer retinal layers (ONL and OSL) show more significant and widespread changes in retinal thickness and correlated most significantly with RPE thickness changes in eyes with GA due to AMD. Assessment of various retinal layer bands can be used as surrogate quantitative parameters to study eyes with GA.

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## [Clinical characteristics, progression and risk factors of geographic atrophy].

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Geographic atrophy (GA) as the late stage manifestation of age-related macular degeneration (AMD) is a progressive disease process afflicting the retinal pigment epithelium, choriocapillaris and the outer neurosensory retina. GA represents a complex, multifactorial disease governed by the interdependence of genetic, endogenous and exogenous factors. Diagnosis and monitoring of GA progression is largely based on various retinal imaging modalities. After the breakthrough in the treatment of wet AMD GA represents a large clinical challenge. Recent studies have contributed to a better understanding of the pathophysiological pathways, natural history and predictive markers for progression.

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## Automated segmentation of geographic atrophy in fundus autofluorescence images using supervised pixel classification.

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Geographic atrophy (GA) is a manifestation of the advanced or late stage of age-related macular degeneration (AMD). AMD is the leading cause of blindness in people over the age of 65 in the western world. The purpose of this study is to develop a fully automated supervised pixel classification approach for segmenting GA, including uni- and multifocal patches in fundus autofluorescence (FAF) images. The image features include region-wise intensity measures, gray-level co-occurrence matrix measures, and Gaussian filter banks. A [Formula: see text]-nearest-neighbor pixel classifier is applied to obtain a GA probability map, representing the likelihood that the image pixel belongs to GA. Sixteen randomly chosen

FAF images were obtained from 16 subjects with GA. The algorithm-defined GA regions are compared with manual delineation performed by a certified image reading center grader. Eight-fold cross-validation is applied to evaluate the algorithm performance. The mean overlap ratio (OR), area correlation (Pearson's [Formula: see text]), accuracy (ACC), true positive rate (TPR), specificity (SPC), positive predictive value (PPV), and false discovery rate (FDR) between the algorithm- and manually defined GA regions are [Formula: see text], [Formula: see text], [Formula: see text], [Formula: see text], [Formula: see text], [Formula: see text], and [Formula: see text], respectively.

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## Long Term Time-Lapse Imaging of Geographic Atrophy: A Pilot Study.

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Geographic atrophy (GA), the late stage of age-related macular degeneration, is a major cause of visual disability whose pathophysiology remains largely unknown. Modern fundus imaging and histology revealed the complexity of the cellular changes that accompanies atrophy. Documenting the activity of the disease in the margins of atrophy, where the transition from health to disease occurs, would contribute to a better understanding of the progression of GA. Time-lapse imaging facilitates the identification of structural continuities in changing environments. In this retrospective pilot study, we documented the long-term changes in atrophy margins by time-lapse imaging of infrared scanning laser ophthalmoscopy (SLO) and optical coherence tomography (OCT) images in 6 cases of GA covering a mean period of 32.8 months (range, 18-72). The mean interval between imaging sessions was 2.4 months (range, 1.4-3.8). By viewing time-lapse sequences we observed extensive changes in the pattern of marginal hyperreflective spots, which associated fragmentation, increase and/or disappearance. Over the entire span of the follow-up, the most striking changes were those affecting hyperreflective spots closest to margins of atrophy, on the non-atrophic side of the retina; a continuum between the successive positions of some of the hyperreflective spots was detected, both by SLO and OCT. This continuum in their successive positions resulted in a subjective impression of a centrifugal motion of hyperreflective spots ahead of atrophy progression. Such mobilization of hyperreflective spots was detected up to several hundred microns away from atrophic borders. Such process is likely to reflect the inflammatory and degenerative process underlying GA progression and hence deserves further investigations. These results highlight the interest of multimodal time-lapse imaging to document cell-scale dynamics during progression of GA. clinicaltrials.gov, identifier: NCT04128150 and NCT04129021.

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## Longitudinal Evaluation of the Development of Wedge-Shaped Subretinal Hyporefectivity in Geographic Atrophy.

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The authors present the case of a 73-year-old woman affected by geographic atrophy (GA) in her left eye. During her follow-up, she developed a subretinal hyporefective space visible on optical coherence tomography (OCT) images. This peculiar OCT sign was recognized to represent a wedge-shaped subretinal hyporefectivity (WSSH). In contrast to previous reports showing that WSSH develops during the flattening of drusenoid pigment epithelium detachment (PED), the authors found that the WSSH manifestation may be even consecutive to the regression of a large PED and consequent GA appearance.

Therefore, the origin of this peculiar OCT finding could be more complex and variegated. [Ophthalmic Surg Lasers Imaging Retina. 2020;51:116-118.].

## Natural History of Visual Acuity and Microperimetry-Based Functional Outcome Measures of the Macula in Patients with Geographic Atrophy: A Retrospective Chart Review Study in Germany.

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**Background/Objectives:** Geographic atrophy (GA) is an advanced form of age-related macular degeneration (AMD) leading to the progressive and irreversible loss of visual function. Characteristics of GA include atrophic lesions resulting from the loss of photoreceptors, retinal pigment epithelium, and choriocapillaris. During GA progression, atrophic lesions typically advance from the macular periphery to the center, affecting foveal light sensitivity and visual acuity. This study analyzed changes in light sensitivity and visual acuity during the natural course of GA progression using the topographic analysis of structural and functional changes based on Early Treatment Diabetic Retinopathy Study (ETDRS) charts, multimodal imaging, and microperimetry assessment. **Methods:** Medical chart data of GA patients between 2014 and 2022 from the Internationale Innovative Ophthalmochirurgie GbR (I.I.O.) research center (Düsseldorf, Germany) were retrospectively analyzed. All patient eyes fulfilling the phase 3 OAKS study inclusion criteria were included and followed up for 60 months. The imputation of missing measurements and dropouts was performed by linear mixed models. **Results:** A total of 20 GA eyes from 13 GA patients were included in the study. At the index, 53.8% of patients had bilateral GA, with 70.0% of the eyes showing multifocal GA and 30.0% subfoveal encroachment (SFE). A total of 35.0% of the eyes had 2-5, and 15.0% over 20, areas of atrophy. Over time, the GA lesion size increased from 6.4 mm<sup>2</sup> to 11.8 mm<sup>2</sup> (1.08 mm<sup>2</sup>/year). After an average observation time of 2.9 years, 78.6% of the initially unaffected study eyes developed SFE. The percentage of study eyes without visual impairment decreased from 55.0% to 30.0%, with mean normal-luminance best-corrected visual acuity (NL-BCVA) reducing from 63.7 to 55.7 ETDRS letters. The share of absolute scotoma points in microperimetry assessment increased from 15.7% to 43.5% while overall average macular sensitivity declined from 15.7 dB to 7.4 dB. **Conclusions:** The substantial deterioration of macular outcomes and visual function was comprehensively detected. The results were a documentation of structural and functional aspects of the natural progression of GA for a 60-month follow-up, providing a typical outline for AMD patients with GA.

## A hybrid segmentation approach for geographic atrophy in fundus auto-fluorescence images for diagnosis of age-related macular degeneration.

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Fundus auto-fluorescence (FAF) images with hypo-fluorescence indicate geographic atrophy (GA) of the retinal pigment epithelium (RPE) in age-related macular degeneration (AMD). Manual quantification of GA is time consuming and prone to inter- and intra-observer variability. Automatic quantification is important for determining disease progression and facilitating clinical diagnosis of AMD. In this paper we describe a hybrid segmentation method for GA quantification by identifying hypo-fluorescent GA regions from other interfering retinal vessel structures. First, we employ background illumination correction exploiting a non-linear adaptive smoothing operator. Then, we use the level set framework to perform segmentation

of hypo-fluorescent areas. Finally, we present an energy function combining morphological scale-space analysis with a geometric model-based approach to perform segmentation refinement of false positive hypo-fluorescent areas due to interfering retinal structures. The clinically apparent areas of hypo-fluorescence were drawn by an expert grader and compared on a pixel by pixel basis to our segmentation results. The mean sensitivity and specificity of the ROC analysis were 0.89 and 0.98%.

## Small Hyperreflective Retinal Foci as in vivo imaging feature of resident microglia activation in geographic atrophy.

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Geographic atrophy (GA), the atrophic late stage of age-related macular degeneration (AMD), is one of the leading causes of vision loss in developed countries. Based on genetic, histological and preclinical studies, the role of the innate immune system in the development and progression of GA is well established. Microglia, the principal resident immune cells, are recognized as key players in innate immunity and contributors to AMD development. Optical coherence tomography (OCT) allows to identify small hyperreflective retinal foci (HRF) with specific features known as aggregates of activated microglial cells as possible in vivo imaging feature of local neuroretinal inflammation. The purpose of this study was to evaluate the presence and amount of small HRF in the eyes of patients with different macular atrophic phenotypes. Patients with GA in both eyes (bilateral GA: B-GA group), patients with GA in one eye and macular new vessels (MNV) in the fellow-eye (unilateral GA: U-GA group) and patients with extensive macular atrophy with pseudodrusen (EMAP), a rare and aggressive variant of atrophic AMD, were retrospectively analyzed. HRF, defined as isolated punctiform elements of small dimensions ( $\leq 30 \mu\text{m}$ ) with intermediate reflectivity (similar to that of the nerve fiber layer) and without a shadow cone, were manually identified and quantified. The amount of HRF was correlated to best corrected visual acuity (BCVA), GA lesion size, measured both at near infrared reflectance (NIR), and blue wavelength fundus autofluorescence (FAF) images, to some GA features (multifocal versus unifocal GA; presence versus absence of foveal sparing) and to central retinal thickness (CRT). Forty-six patients (26 in the B-GA group, 16 in the U-GA group and 4 in the EMAP group) were studied. Patients with EMAP were younger compared to patients with B-GA and to patients with U-GA ( $63.5 \pm 6.8$  years vs  $80.4 \pm 8.4$  years B-GA, and vs  $83.3 \pm 6.1$  years U-GA;  $p = 0.0004$  and  $p = <0.0001$ , respectively). Mean BCVA, mean GA area at NIR and at FAF images, foveal sparing and multifocal versus unifocal GA distribution and mean CRT were not significantly different among groups. GA area was wider on NIR versus FAF in all groups, significantly in B-GA and U-GA groups ( $11.7 \pm 7.6 \text{ mm}^2$  vs  $10.6 \pm 7.1 \text{ mm}^2$ ,  $p = 0.0087$  in B-GA;  $7.8 \pm 9.2 \text{ mm}^2$  vs  $7.7 \pm 9.4 \text{ mm}^2$ ,  $p = 0.004$  in U-GA). The number of HRF was significantly higher in U-GA compared to B-GA and to EMAP ( $47.4 \pm 7.1$  vs  $31.6 \pm 7.3$  B-GA and  $28.0 \pm 4.9$  EMAP,  $p < 0.0001$  for both), while mean HRF number did not significantly differ between B-GA and EMAP ( $p = 0.1960$ ). HRF count correlated only to CRT, positively in B-GA and negatively in U-GA group. The increase of small HRF, which mirrors retinal microglial activation, characterizes eyes with unilateral GA (and MNV in the fellow eye) but not eyes with bilateral GA or EMAP. The role of activated microglia in the retina of GA eyes needs to be better investigated, mainly considering the actual and new therapeutic strategies with which to reduce either the development or progression of the atrophic macular changes.

## Epidemiology of geographic atrophy and its precursor features of intermediate

## age-related macular degeneration.

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Globally age-related macular degeneration (AMD) is a leading cause of blindness with a significant impact on quality of life. Geographic atrophy (GA) is the atrophic late form of AMD and its prevalence increases markedly with age with around 1 in 5 persons aged 85 and above having GA in at least one eye. Bilateral GA leads to severe visual impairment thus posing a significant burden on patients, careers and health providers. The incidence and prevalence of GA varies across different geographic regions, with the highest rates in those of European ancestry. Although heterogeneity in definitions of GA and reporting strategy can explain some of the discrepancies, the data overall are consistent in showing a lower prevalence in other ethnicities such as those of Asian heritage. This is at present unexplained but thought to be due to the existence of protective factors such as differences in eye pigmentation, diet, environmental exposures and genetic variability. This review covers key aspects of the prevalence and incidence of the ocular precursor features of GA (large drusen, pigmentary abnormalities and reticular pseudo-drusen), the late stage of GA and factors that have been known to be associated with modifying risk including systemic, demographic, environment, genetic and ocular. Understanding the global epidemiology scenario is crucial for the prevention of and management of patients with GA.

## Imaging geographic atrophy: integrating structure and function to better understand the effects of new treatments.

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Geographic atrophy (GA) is an advanced and irreversible form of age-related macular degeneration (AMD). Chronic low grade inflammation is thought to act as an initiator of this degenerative process, resulting in loss of photoreceptors (PRs), retinal pigment epithelium (RPE) and the underlying choriocapillaris. This review examined the challenges of clinical trials to date which have sought to treat GA, with particular reference to the successful outcome of C3 complement inhibition. Currently, optical coherence tomography (OCT) seems to be the most suitable method to detect GA and monitor the effect of treatment. In addition, the merits of using novel anatomical endpoints in detecting GA expansion are discussed. Although best-corrected visual acuity is commonly used to monitor disease in GA, other tests to determine visual function are explored. Although not widely available, microperimetry enables quantification of retinal sensitivity (RS) and macular fixation behaviour related to fundus characteristics. There is a spatial correlation between OCT/fundus autofluorescence evaluation of PR damage outside the area of RPE loss and RS on microperimetry, showing important associations with visual function. Standardisation of testing by microperimetry is necessary to enable this modality to detect AMD progression. Artificial intelligence (AI) analysis has shown PR layers integrity precedes and exceeds GA loss. Loss of the ellipsoid zone has been recognised as a primary outcome parameter in therapeutic trials for GA. The integrity of the PR layers imaged by OCT at baseline has been shown to be an important prognostic indicator. AI has the potential to be invaluable in personalising care and justifying treatment intervention.